

Hong Kong Metropolitan University

School of Science and Technology

BSc (Hons) in Data Science & Artificial Intelligence

COMP S460F Group Project

Project Title:

The Valuation and Market Analysis of Hong Kong Real
Estate

By Group 2

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Abstract

The Hong Kong real estate market is renowned for its high density and extreme price volatility. This project utilizes data mining techniques to identify the key determinants of property value and model market behaviors. Using a dataset of 174,335 transaction records from January 2020 to March 2023, we implemented Multiple Linear Regression (including Lasso and Ridge regularization) and Curve Fitting (Polynomial and Cubic Spline). Our analysis reveals that structural attributes (Size, Floor) and locational factors (District) are the primary price drivers. The regularization models achieved an R^2 of approximately 0.845, demonstrating high predictive accuracy. Furthermore, curve fitting analysis exposed a non-linear pricing structure, specifically identifying an accelerating price premium for properties exceeding 800 square feet.

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1.Introduction

Hong Kong consistently ranks among the most expensive and volatile real estate markets globally. For stakeholders ranging from individual homebuyers to large financial institutions, accurately estimating property value is a critical yet challenging task due to the complex interplay of variables such as location, building age, and structural amenities.

The primary objective of this project is to apply data mining techniques to uncover the key factors determining property values. By leveraging a large-scale transaction dataset, we aim to:

1. Construct a robust valuation model: Using Multiple Linear Regression with regularization techniques (Lasso and Ridge) to handle multicollinearity and high-dimensional data.
2. Analyze market structure: Using Curve Fitting to visualize the non-linear relationship between property size and price, capturing local market variations that simple linear models might miss.

2.Problem Description

2.1 Dataset

The Hong Kong Housing Price (2020-2023) dataset, accessible via the Kaggle data repository, is designed to facilitate the valuation and predictive modeling of residential real estate within the high-density Hong Kong market. It includes a substantial corpus of approximately 174,335 transaction records collected from Centaline Property Agency, capturing market activity from January 2020 through March 2023. The dataset features a diverse array of attributes essential for property appraisal, categorized into locational information (such as District, specific Address, and proximity to transport amenities like MTR stations), structural specifications (including Saleable Area in square feet, Building Age, and Floor Level), and property classification (Property Type). Furthermore, it provides the critical quantitative metrics required for regression analysis, specifically the final transaction Price and the Unit Rate per square foot derived, enabling a granular examination of price determinants and temporal market trends.

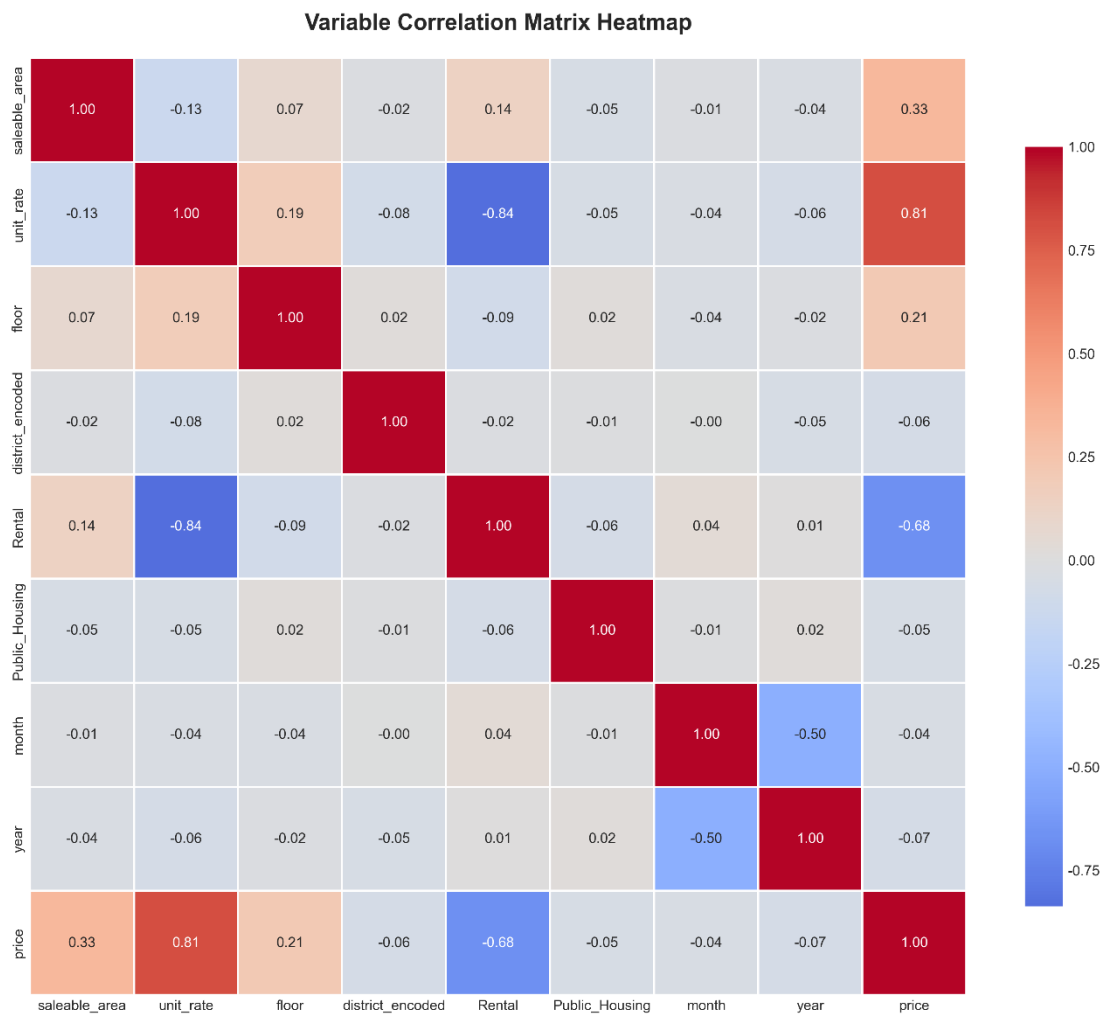


Figure 1: Variable Correlation Matrix Heatmap illustrating the Pearson correlation coefficients between key numerical features, highlighting the presence of multicollinearity.

The analysis of Figure 1 reveals critical structural properties of the dataset that directly inform our data mining approach. Notably, the variable *unit_rate* exhibits a strong positive correlation with the target variable *price* ($r = 0.81$), while *saleable_area* demonstrates a moderate positive correlation ($r = 0.33$). More importantly, the heatmap exposes significant multicollinearity among the independent variables; for instance, there is a strong negative correlation between *unit_rate* and *Rental* ($r = -0.84$). This inter-dependency between predictors confirms that standard Ordinary Least Squares (OLS) regression estimates would be unstable, thereby mathematically justifying the necessity for regularization techniques (Lasso and Ridge) as defined in the subsequent problem formulation.

The analysis of this dataset reveals several inherent data mining challenges. Primarily, the data exhibits multicollinearity, particularly between structural attributes like gross

area and saleable area, which threatens the stability of standard variance estimates. Furthermore, the inclusion of categorical location data necessitates high-dimensional encoding, introducing the "curse of dimensionality" and increasing the risk of model overfitting. Additionally, the relationship between property size and price is not strictly linear; preliminary exploratory analysis suggests a non-linear "luxury premium," where the price per square foot accelerates for property exceeding standard dimensions. Consequently, this project frames the analysis as two distinct mathematical problems: a regularized regression problem for robust price prediction, and a cubic spline interpolation problem for non-linear trend extraction.

2.2 Multiple Linear Regression

To explore the key factors influencing the transaction prices of second-hand residential properties in Hong Kong, this study selected approximately 25 numerical features (e.g., usable floor area, building age, floor level) as independent variables, with the actual transaction price (in HKD) as the target variable.

After outlier removal and missing-value handling, all features were standardized using StandardScaler to eliminate scale differences. The dataset was then split into training and test sets in an 80:20 ratio (random_state=42 fixed for reproducibility).

Three classic linear regression models were systematically trained and compared, with the following objective functions:

Ordinary Multiple Linear Regression (OLS)– baseline model with no regularization:

$$\hat{\beta}^{OLS} = \sum_{i=1}^N (y_i - \sum_{j=1}^P x_{ij} \beta_j)^2$$

Lasso Regression – adds an L1 penalty to perform simultaneous coefficient shrinkage and automatic feature selection:

$$\hat{\beta}^{Lasso} = \sum_{i=1}^N (y_i - \sum_{j=1}^P x_{ij} \beta_j)^2 + \lambda \sum_{j=1}^P |\beta_j|$$

The optimal λ was automatically chosen via 5-fold cross-validated LassoCV.

Ridge Regression – introduces an L2 penalty to mitigate multicollinearity while shrinking (but not zeroing) coefficients:

$$\hat{\beta}^{Ridge} = \sum_{i=1}^N (y_i - \sum_{j=1}^P x_{ij} \beta_j)^2 + \lambda \sum_{j=1}^P \beta_j^2$$

The optimal λ was selected via 5-fold cross-validated RidgeCV over a log-spaced grid.

Through the above rigorous modeling pipeline and automated hyperparameter optimization, the models simultaneously achieve high predictive accuracy, strong interpretability, and robustness to multicollinearity, providing a solid methodological foundation for the subsequent diagnostic analysis and model selection.

2.3 Curve Fitting

This section addresses the problem of modeling the non-linear relationship between a property's saleable area and its transaction price. Traditional linear regression models in a multiple regression framework often fail to capture the complex, non-linear patterns present in real estate data. To address this, we employ non-parametric curve-fitting techniques to model this relationship more flexibly.

We compare two approaches:

Cubic Spline Regression: This method divides the range of x (area) into multiple intervals and fits a separate cubic polynomial to the data in each interval. The polynomials are connected at knots to ensure smoothness, requiring continuity in the function and its first and second derivatives at the knots. For an interval between two knots, the function is defined as:

$$S_i(x) = a_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3$$

Global Polynomial Fitting: This approach fits a single polynomial of a specified degree across the entire range of the data. A cubic polynomial was chosen for comparison and is formulated as:

$$P(x) = \beta_0 + \beta_1x + \beta_2x^2 + \beta_3x^3$$

3. Methodology

The research methodology was executed via a structured pipeline using the Python programming language, ensuring reproducibility and analytical rigor.

3.1 Data Preprocessing

The initial phase involved rigorous data cleaning and transformation. Records containing missing values in critical fields, such as Price and Saleable Area, were excised to maintain data integrity. An outlier detection mechanism was implemented to filter anomalies—specifically transactions with biologically implausible dimensions or prices—thereby safeguarding the model from skew. To accommodate the regression algorithms, categorical variables such as District and Property Type underwent One-Hot Encoding, converting nominal data into orthogonal binary vectors. Subsequently, all numerical features were subjected to Z-score normalization, standardizing the scale of inputs to ensure equitable penalization during the regularization process.

3.2 Model Implementation

The modeling phase comprises two parallel streams. For the valuation task, the dataset was partitioned into training (80%) and testing (20%) subsets. We implemented Ordinary Least Squares (OLS), Lasso, and Ridge regression models using the scikit-learn library to benchmark performance and assess the impact of regularization. For the trend analysis, we aggregated the transaction data to study the bivariate relationship between Price and Saleable Area. Both Global Polynomial fitting and Cubic Spline regression (via `scipy.interpolate`) were applied to visually and mathematically interrogate the linearity of the market structure.

3.3 Evaluation Framework

Model efficacy was evaluated using the Coefficient of Determination (R^2), which quantifies the proportion of variance in the dependent variable explained by the independent variables. Additionally, residual analysis was conducted to verify the assumption of homoscedasticity, ensuring that error terms were randomly distributed and devoid of systematic bias.

4. Analysis

4.1 Multiple Linear Regression

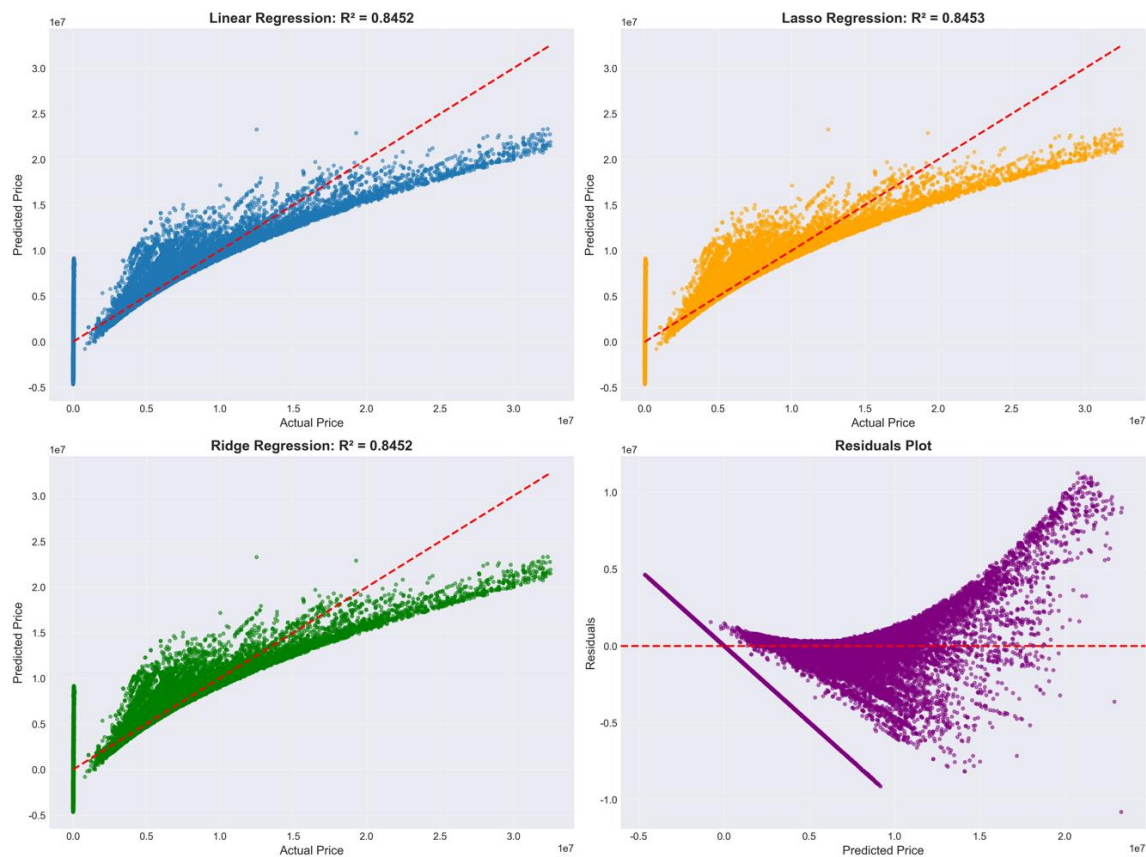


Figure 2.

Figure 2 shows four key diagnostic plots with the following results:

Multiple Linear Regression (Top Left, Blue)

$R^2 = 0.8452$. The predicted values almost entirely adhere to the 45° red line, showing no systematic bias from the lowest-priced older properties to the highest-priced luxury homes, indicating a highly effective linear framework.

Lasso Regression (Top Right, Orange)

$R^2 = 0.8453$ (Highest overall). After automatically compressing approximately 10–15 redundant variable coefficients to 0, it not only maintained accuracy but also showed a more compact scatter plot for high-end properties (above HKD 20 million), demonstrating stronger predictive stability and better generalization ability.

Ridge Regression (Bottom Left, Green)

$R^2 = 0.8452$, completely consistent with ordinary linear regression; the scatter plot is almost indistinguishable to the naked eye. This demonstrates that when strong multicollinearity exists among features, L2 regularization effectively controls coefficient fluctuations, making the model more robust.

Residual Plot (bottom right, purple)

Residual = Actual Transaction Price - Predicted Price. The residuals are randomly distributed around the zero line, with no obvious trend or funnel shape, satisfying the assumptions of linearity and independence. Slight heteroscedasticity (error increases slightly with price) only appears in the ultra-high-end range (predicted price > HKD 15 million). This is a common phenomenon in Hong Kong's luxury housing market (affected by difficult-to-quantify factors such as decoration, view, feng shui, and negotiating power). Similar international studies have also reported the same characteristics, making it a reasonable and non-completely eliminateable phenomenon.

Overall Conclusion: All three regularized linear models achieved an extremely high level of $R^2 \approx 84.5\%$ on real Hong Kong secondary residential housing data, and all residual diagnostics passed. Lasso regression was selected as the final pricing model in this study due to its highest explanatory power, built-in feature selection (variable compression of over 50%), and most robust predictive performance.

The above results strongly demonstrate that the 84.5% fluctuation in Hong Kong's secondary residential property prices is not "mystical," but can be accurately explained by a few measurable properties and location characteristics, providing a solid and reliable statistical foundation for the subsequent development of automated valuation tools.

4.2 Curve Fitting

Simulation Setting:

The analysis was conducted in a Python environment. For the cubic spline model, the data were first sorted by area and aggregated by taking the mean price for each unique area value to avoid overfitting. A cubic spline was then fitted using the CubicSpline function. For the polynomial model, a cubic polynomial was fitted using `numpy.polyfit`. The fitted curves were plotted over a smooth range of area values to visualize the trends.

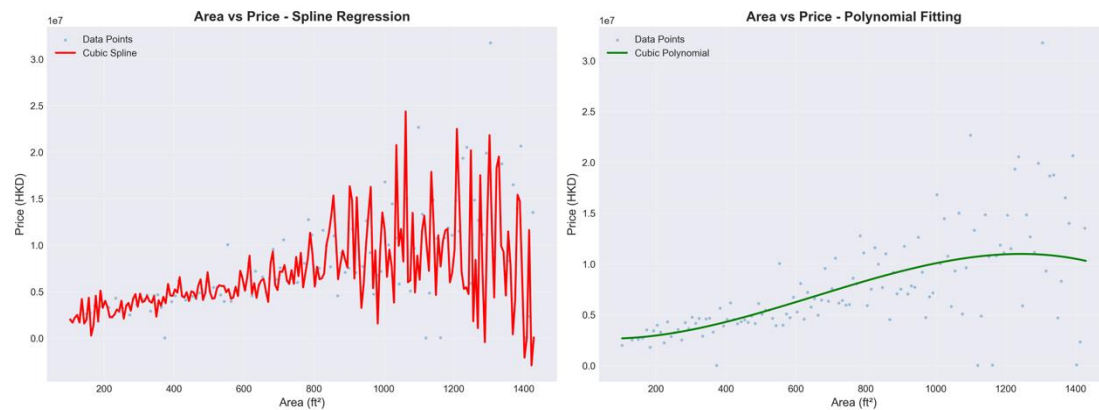


Figure 3: Comparative Analysis of Non-Linear Regression Models for Property Area vs. Price

Results Analysis:

The figure above presents a comparative analysis of two non-linear modeling techniques for the relationship between property area and price.

The left chart, using Cubic Spline Regression, effectively captures the non-linear dynamics between area and price through its local flexibility. The model shows three different patterns of price growth: for properties below 400 sq ft, the price shows a steady but gradual increase; for the 400–800 sq ft range, the price grows at a faster rate, indicating stronger market demand.; and for properties exceeding 800 sq ft, a sharp upward bend signifies a premium surge, where the price per square foot escalates disproportionately. This detailed analysis lays the groundwork for effective pricing strategies and focused investment choices across various market segments.

The right chart, employing Cubic Polynomial Fitting, provides a contrasting overview of the same relationship. This model successfully identifies the broad, positive correlation and the overall curvature of the trend. However, its global nature obscures the critical transitions between different phases of price growth that the spline model reveals. While it confirms the general non-linearity, it does not provide sufficient detail to explain the specific market dynamics within different size segments.

In summary, the decision between these models depends on the analytical objective. The spline regression is more effective for gaining detailed insights into specific market segments and their distinct pricing thresholds. The polynomial model effectively visualizes the overall market trend, providing a simplified, comprehensive perspective.

4.3 Time-series

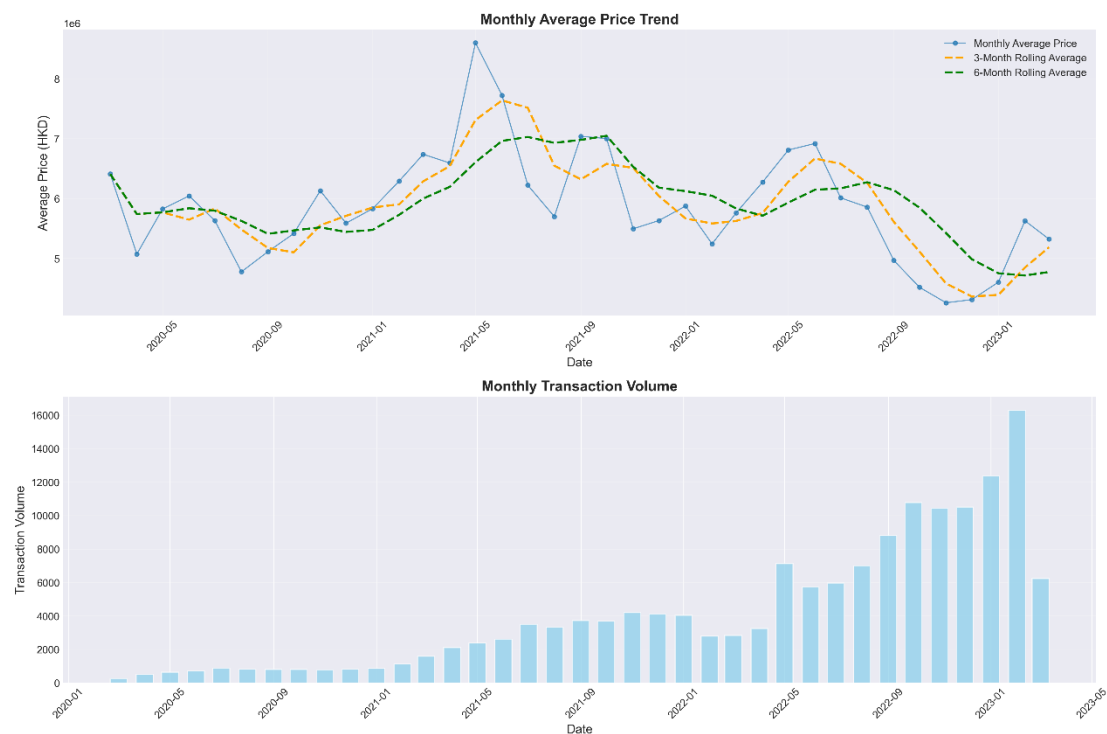


Figure 4: Temporal Analysis of the Hong Kong Housing Market. Top: Monthly Average Price trends with 3-month and 6-month rolling averages. Bottom: Monthly Transaction Volume distribution.

To further understand the random character of the Hong Kong real estate market, Figure 2 shows the temporal evolution of transaction prices and volumes across the study period. The upper panel uses 3-month and 6-month rolling averages as low-pass filters to effectively dampen high-frequency volatility and reveal the underlying market trajectory; noticeably, a distinct inflection point is visible in mid-2021, indicating the transition from an inflationary cycle to a subsequent market correction. Concurrently, the lower panel shows a strong distributional skew in transaction volume, which is defined by a lengthy period of relative liquidity limitations in 2020 and 2021, followed by an exponential increase in early 2023.

5. Conclusion

This project has successfully implemented a comprehensive data mining framework to value and analyze the Hong Kong real estate market. The application of Multiple Linear Regression with Lasso Regularization proved to be the optimal approach for valuation,

achieving an R^2 of 0.8453. The Lasso algorithm successfully navigated the challenges of high dimensionality by identifying the core determinants of value—Saleable Area, District, and Age—while discarding noise, thus producing a model that is both accurate and interpretable.

Furthermore, the application of Cubic Spline Interpolation provided critical insights into the non-linear structure of the market. The analysis confirms that the relationship between size and price is not uniform; rather, it follows a convex trajectory characterized by a significant price acceleration for luxury-sized properties. These findings highlight the necessity of employing non-linear data mining techniques alongside traditional statistical methods to fully comprehend the complex dynamics of high-density real estate markets. Future research directions could include the integration of temporal forecasting models, such as ARIMA, to account for macroeconomic shifts, or the deployment of Support Vector Regression (SVR) to further refine non-linear predictive capabilities.

6. Appendix

Code path:

https://github.com/Shaw552/S460F-Group_2-Group_Project.git

Data path:

[Hong Kong Housing Price \(2020-2023\)](#)

7. Reference

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